Amr Abdallah

Canada
relatora@gmail.com • LinkedIn, GitHub, Website

Applied AI Engineer with 10+ years of software engineering, putting AI agents and LLM-powered applications into production. Delivered a Microsoft Copilot Studio agent in Teams and Copilot that answers stakeholder questions about quality, purchasing, and KPI data, and partnered with Monolith AI on predictive models for experiments and chemical reactions. Brings a solution architect's view to AI work, designing the secured APIs, cloud infrastructure, and data pipelines that let models reach enterprise data safely.

Core Competencies

- **AI & Agentic Systems**: AI agents, LLM integration, RAG, tool/function calling, Model Context Protocol (MCP), prompt engineering, Microsoft Copilot Studio, multi-agent coding with Claude Code, local LLMs, predictive modeling.
- **Full-Stack Engineering**: MERN (MongoDB, Express, React, Node.js), Go, Python (FastAPI/Flask), Java (Spring), .NET Core/C#, PHP (Laravel), Vue.js, TypeScript, REST APIs, GraphQL.
- **Solutions Architecture**: Microservices, distributed and event-driven systems, real-time IoT (MQTT, Modbus, WebSocket), secured API design, cloud infrastructure design.
- **Azure**: Functions, App Service, Container Apps, Virtual Machines, Cosmos DB, Blob Storage, SignalR, API Management, Entra ID, Azure Data Explorer, Power BI.
- **AWS**: Lambda, API Gateway, EC2, Elastic Beanstalk, container services, Amplify, DynamoDB, MemoryDB, S3, EFS, IAM, CloudWatch, CloudTrail, Network Firewall, AWS Backup.
- **Data Systems**: PostgreSQL, MySQL, Oracle, DB2, MS SQL, MongoDB, Kusto, Grafana, Redis, Supabase, vector databases.
- **DevOps & CI/CD**: Docker, Terraform, CI/CD pipeline design with Jenkins and GitHub webhooks, Gitea self-hosted on the internal network.
- **Technical Leadership & Delivery**: Full-lifecycle delivery, requirements, specifications and Visio diagrams, backlog prioritization, Agile/Scrum (Jira, Azure DevOps), cross-functional team leadership.

Professional Experience

- **AlumaPower**
  Senior Software Engineering Lead *2022 - Present*
  - Delivered a Copilot Studio agent running on Claude, used company-wide in Teams (thousands of questions a month), that answers quality, purchasing, and KPI questions and flags recurring failures
  - Implemented the secured, read-only Azure API behind the agent (Entra ID), exposing curated MongoDB views as LLM tools that match QMS records with NCRs and ECRs
  - Created the data-cleaning pipeline feeding Monolith AI's platform, whose predictive models for experiments and chemical reactions replaced many physical tests and cut costs by 32%
  - Built the company's analytics platform, turning operational logs and test-technician experiment data into the KPIs behind company decisions and reports, with Docker-hosted Redis caching large queries
  - Engineered a real-time IoT data pipeline: hardware boards publish MQTT messages to Azure Web Apps that decode and normalize multi-sensor readings onto shared timestamps, feeding Kusto (Grafana dashboards), MongoDB, and a live client that charts telemetry and starts/stops fans and motors
  - Architected a company-wide MERN suite behind one authentication gateway, spanning QMS, SCM, HR, assets, quality data, fabrication, and ticketing, saving thousands of hours
  - Set up CI/CD with Jenkins and GitHub webhooks for all web apps, Terraform provisioning, and a self-hosted Gitea server on the internal network; own unit, regression, and end-to-end testing
  - Lead the engineering team, owning hiring, performance reviews, and delivery with leadership and partner teams
  - Developed a C# WinForms data logger and controller that charts live voltage, current, temperature, and flow and actuates hardware; now migrating it to React and .NET Core

- **Industrial Web Apps**
  IT Project Manager *2021 - 2022*
  - Contributed to SHORE, the company's end-to-end project management platform, built with React, Node.js, and PostgreSQL on AWS (API Gateway, EC2, Elastic Beanstalk, S3, IAM, CloudTrail)
  - Directed cross-functional teams across the full project lifecycle, maintained the backlog, and prioritized work across simultaneous development and deployment projects
  - Set up Docker-based development environments and web deployments
- **Lambton College**
  Professor (Part-Time) *2021 - Present*
  - Teach Front-End Web Development I (HTML5, CSS3) and II (JavaScript, jQuery, Vue, React), and supervise co-op students
- **Eye-Look Media**
  Full-Stack Developer *2020 - 2021*
  - Authored most of the Laravel backend and the Electron content editor of a digital signage portal (Laravel, Vue, MySQL) for clients such as Canadian Tire
  - Developed a client-facing progressive web app over WebSocket, plus Node.js microservices for automation
- **Avaros EMR**
  Software Developer (Co-op) *2020*
  - Created a patient online-booking frontend (React, React Hooks, RxJS, TypeScript)
  - Integrated Twilio SMS webhooks into a Go backend with PostgreSQL migrations
- **Lambton College Applied Research**
  Researcher (Contracts) *2019 - 2023*
  - Engineered an IoT solution for Just Vertical: a Flutter app over WebSocket to an AWS Lambda microservice, bridging to a Quasar/Electron (Vue.js) app on Raspberry Pi units for irrigation/lighting automation
  - Delivered a MERN-stack operations system for Procor (railcar operations) and a Raspberry Pi digital signage microservice solution for Eye-Look Media
- **Earlier Professional Experience**
  Software Developer and Team Lead, then ICT Teacher
  - Delivered an SMS-based banking solution for a national bank and a Pocket PC sales-force automation system with MS SQL replication; later designed ICT curricula and school software, including a grading system

Selected Projects

- **AI Painting Estimator** (GitHub): Designed and shipped a progressive web app that turns a photo and job description into a professional painting quote. A vision model (Gemini Flash, behind a swappable adapter) flags surfaces, condition, and prep work; a deterministic pricing engine does the arithmetic, and the contractor confirms every price. React, TypeScript, Hono on Cloudflare Workers, D1, R2.
- **Engineering Work Dashboard** (GitHub): Developed an org-wide dashboard that measures engineering output across every repository and branch of a GitHub organization, filtering generated-file noise that can double raw line counts, categorizing commits by type of work, and exporting PDF reports for leadership. Rate-limit-aware, resumable scanning over the GitHub API. Next.js, TypeScript, PostgreSQL (Drizzle), Docker.

Education

- **Lambton College**
  Diploma, Computer Software & Database Development *2019 - 2021*
  Cumulative GPA 3.89/4.0.
- **6th of October University**
  Bachelor of Business Administration, Finance (minor: Economics)
  Degree: Very Good. Graduation Project: Distinct.